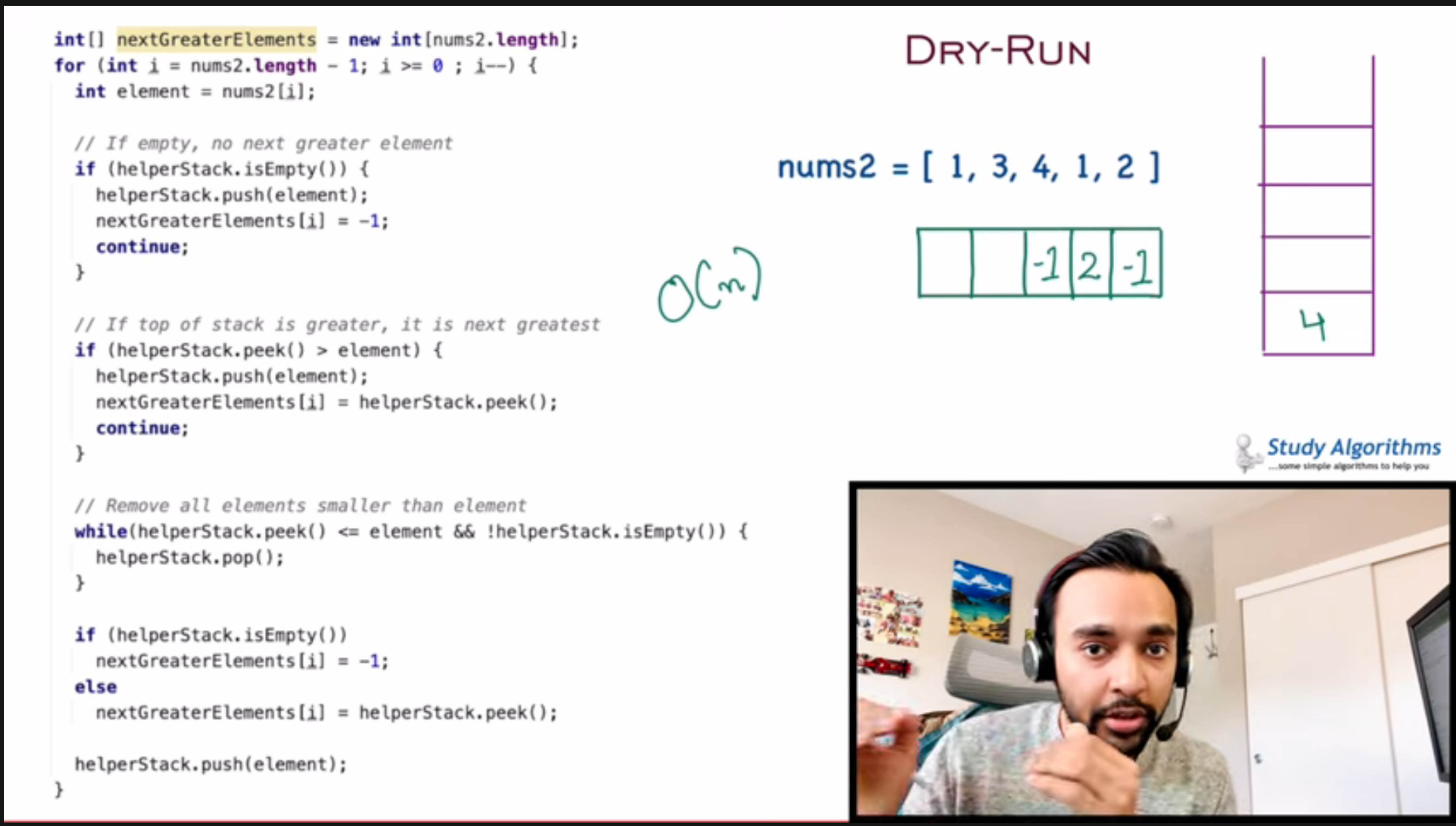
Next Greater Element I (LC 496) * ************************************

LEARN THIS LEARN THIS ****************************** IMPORTANT**



it was next greater element to the right of current element

learn how we created the nge in this , it was nge to the right , so we started from right then to left

LEARN LEARN THIS NGE ONE , rest you can come up with

if empty push and mark nge i = -1

if element < top mark nge i = top and push element

while element >= top , you pop

if empty , nge = -1

else nge i = top

push element

****

503. Next Greater Element II

Solved

Medium Topics Companies

Given a circular integer array `nums` (i.e., the next element of `nums[nums.length - 1]` is `nums[0]`), return *the **next greater number** for every element in* `nums`.

The **next greater number** of a number `x` is the first greater number to its traversing-order next in the array, which means you could search circularly to find its next greater number. If it doesn't exist, return `-1` for this number.

HINT : you just do it twice on the nums

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739. Daily Temperatures

Solved

Medium | Topics | Companies | Hint

Given an array of integers `temperatures` represents the daily temperatures, return *an array* `answer` *such that* `answer[i]` *is the number of days you have to wait after the* i^{th} *day to get a warmer temperature*. If there is no future day for which this is possible, keep `answer[i] == 0` instead.

HINT : push indices to stack , nge = st.top() - i

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1475. Final Prices With a Special Discount in a Shop

Solved

Easy | Topics | Companies | Hint

You are given an integer array `prices` where `prices[i]` is the price of the i^{th} item in a shop.

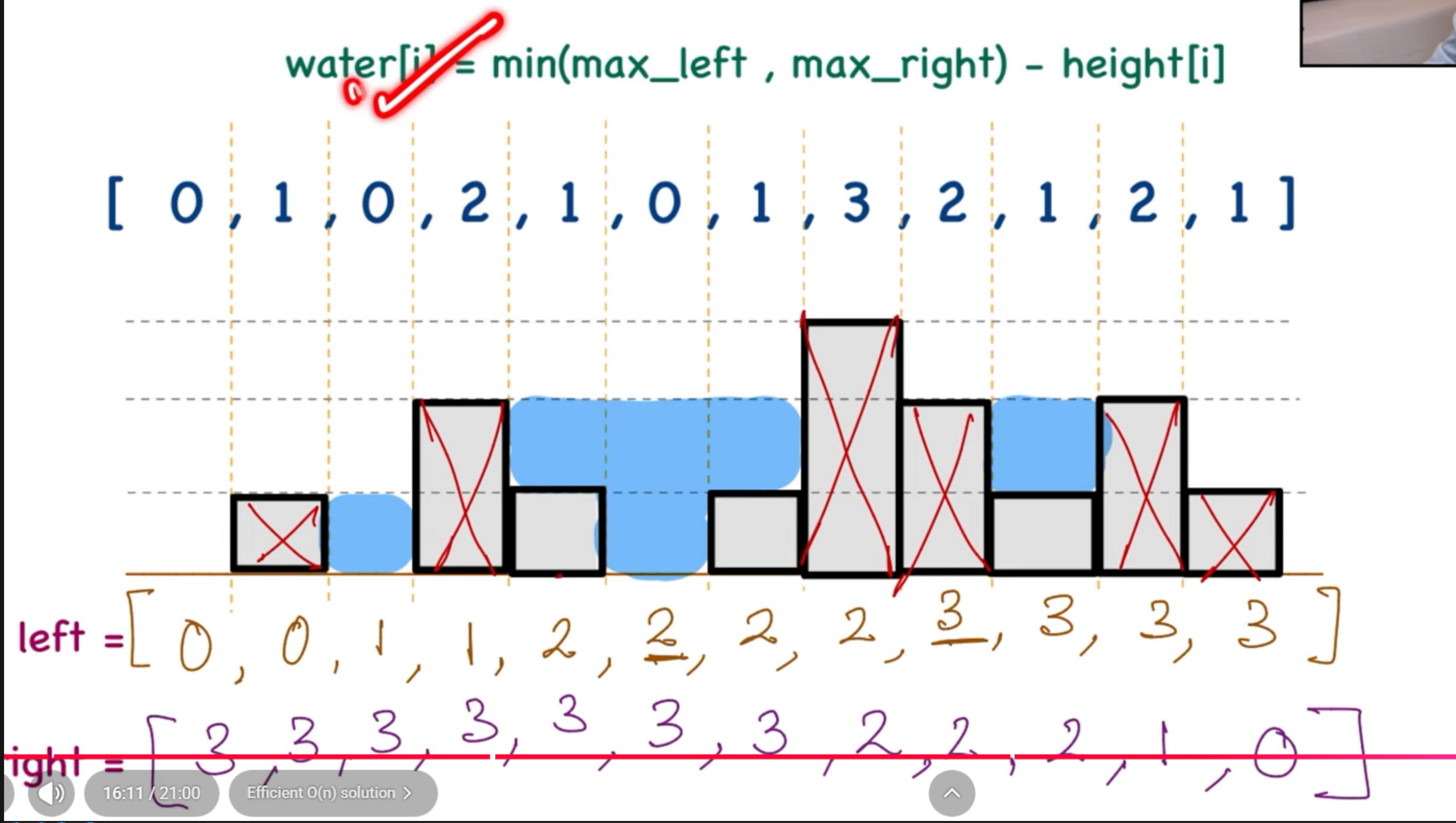
There is a special discount for items in the shop. If you buy the i^{th} item, then you will receive a discount equivalent to `prices[j]` where `j` is the minimum index such that `j > i` and `prices[j] <= prices[i]`. Otherwise, you will not receive any discount at all.

Return an integer array `answer` where `answer[i]` is the final price you will pay for the i^{th} item of the shop, considering the special discount.

ye bhi same hi , NEXT SMALLER ELEMENT implement karna hai , NGE likho and signn ulta kardo

TRAPPING RAIN WATER ***

this solution doesnt uses a stack , we compute both left and right then find the answer , -> 2 scans




```cpp
int trap(vector<int>& height) {
    int n = height.size();

    vector<int> left(n, 0) , right(n, 0);
    int ans = 0 , water ;
    for(int i=1;i<n;i++){
        left[i] = max(height[i-1] , left[i-1]);
    }

    for(int i=n-2;i>=0;i--){
        right[i] = max(height[i+1], right[i+1]);

        water = min(left[i], right[i]) - height[i];
        if(water>0) ans+=water;
    }

    return ans;

}
```